

Budgeted test-time adaptation with rollback for lightweight segmentation

Yuvraj Verma | Proposed first project with Prof. Md Mostafijur Rahman | Texas Tech University

Research pillars: continuous adaptation and reliability; scalable AI for constrained hardware. **Initial settings:** polyp segmentation and aerial building segmentation, trained as separate tasks. **Available platform:** one 6GB RTX 4050 laptop GPU.

1. Motivation and significance

An efficient segmentation model can still fail when illumination, blur, noise or acquisition conditions change. A deployment that repeatedly updates from unlabeled inputs may reduce some of those errors, but it can also become confidently wrong. The first project would ask whether update decisions can be constrained and audited as carefully as the model itself, without introducing a large auxiliary network.

For medical images, a missed or distorted lesion boundary can make a high aggregate overlap misleading. For remote sensing, missing many small roofs can bias an inventory even when large buildings dominate the pixel score. These are different semantic tasks, but both expose a practical engineering question: when should a compact model spend additional computation adapting, and when should it retain its original prediction? This proposal addresses that shared decision, rather than assuming a model trained on one task can recognize the other.

2. Research question and falsifiable hypotheses

Question: under a fixed, measured adaptation budget, can a lightweight accept-or-rollback rule reduce harmful updates while preserving useful gains on shifted inputs? The rule will be evaluated against both frozen inference and the same candidate update without gating.

H1: gating reduces the fraction of cases with a Dice loss exceeding one percentage point relative to the frozen model, compared with an ungated update, at matched candidate-computation budgets. **H2:** an input-level trigger can avoid unnecessary updates on clean cases without losing most of the ungated shifted-case benefit. **H3:** flip consistency alone is an insufficient acceptance test; adding a foreground-change or boundary-localized condition improves its ability to reject damaging updates.

The one-point threshold is a predeclared experimental definition, not a clinical tolerance. Each hypothesis can fail: a gate may reject beneficial candidates, accept consistently wrong masks, or consume more time than the update saves. The project will report those outcomes rather than define success solely through average Dice.

3. Proposed scope

The initial study uses binary segmentation and a small architecture with source-trained BatchNorm. It does not attempt full-model foundation-model tuning, clinical reasoning, multi-organ 3D analysis or cross-task zero-shot transfer. A later brain-MRI study would require an appropriately labeled dataset, subject-level partitions and 3D-aware evaluation; it will not be inferred from these 2D screening results.

Preliminary source and target training are complete; measured results and the small adaptation pilot appear on page 5. The full trigger, boundary localization and repeated-seed study remain proposed work.

Positioning the project and defining the gap

4. What existing methods already address

Tent [3] adapts normalization-related parameters through predictive entropy. EATA [4] selects useful samples and constrains forgetting; CoTTA [5] combines teacher-based predictions with restoration; SAR [6] examines stability under changing test conditions. These works already establish that selective updates, restoration and reliability matter. A rollback mechanism alone is therefore not a novelty claim.

Medical-segmentation-specific work strengthens that caution. On-the-Fly adaptation [7] addresses test distribution differences in medical images. Chen et al. [8] refine foundation-model representations without model parameter updates. MedSeg-TTA [9] compares adaptation paradigms across multiple modalities and shows that their behavior depends on the shift. Recent memory-supported training-free adaptation [10] is another relevant comparison. Methods [8,10] rely on foundation-model components, so they are contextual comparators rather than drop-in MK-UNet algorithms. The primary causal comparison will use the same MK-UNet checkpoint and compute accounting; any foundation-model experiment will report its distinct backbone, prompts and resource budget.

Prior direction	What this study would test
Normalization / entropy adaptation [3]	Does a very small model still make useful candidate updates when its running statistics stay frozen?
Selective / stable adaptation [4-6]	Does a decision rule reduce harmful segmentation updates at a measured compute budget?
Medical / foundation-model approaches [7,8,10]	When are parameter-free or input-level corrections a better use of the available budget?
Cross-modality benchmarking [9]	Can failures be explained by object size, boundaries and shift type instead of a pooled mean?

5. Specific gap and contribution target

The proposed contribution is a controlled segmentation study of the tradeoff between update harm, useful adaptation and total inference cost on a sub-million-parameter backbone. The experiment should make the gate itself a measurable object: false acceptance of harmful updates, false rejection of beneficial updates, spatial localization of unstable predictions, and added latency. Whether this remains a publishable gap will be reassessed against the detailed protocols of [7-10] before claiming novelty.

The project is deliberately narrower than proposing another large adaptable backbone. It should produce an auditable decision policy, an ablation suite, and evidence about when that policy cannot infer correctness from unlabeled observations. If the simple gate fails, identifying the failure boundary remains a useful result and will guide a smaller follow-up.

6. Mathematical problem

Let $f(\theta, x)$ return binary foreground probabilities; θ_0 is a source-trained snapshot. A test case x arrives without its mask y . A candidate θ_1 is formed from x using a restricted parameter subset. A policy g observes unlabeled predictions, input statistics and computational budget, then chooses θ_1 or θ_0 . Ground-truth y is available only to the offline evaluator. The objective is to improve held-out overlap while limiting update harm and cost; label-free proxy improvement is not equivalent to segmentation improvement.

A small candidate update with an explicit decision rule

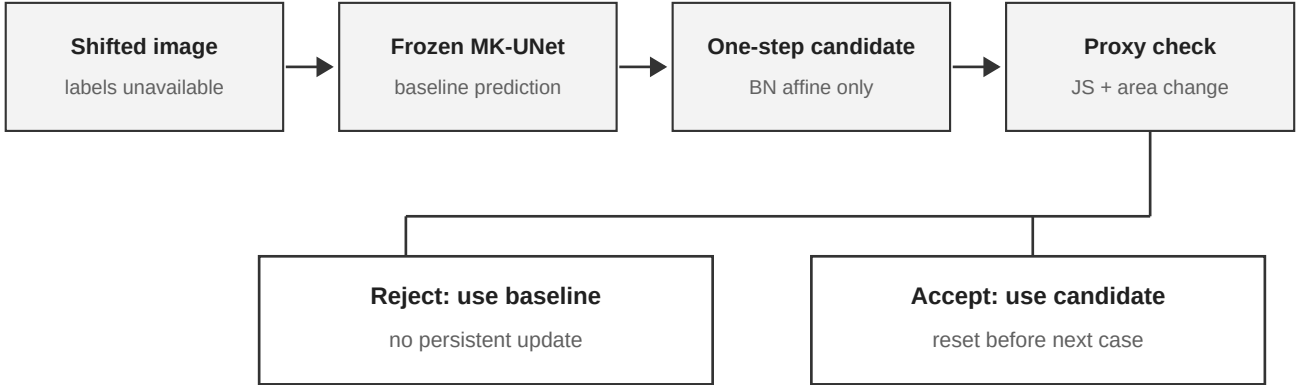


Figure 1. Initial episodic framework. The candidate and acceptance test use no target labels. Accepted weights apply only to the current case; the next case starts from the source snapshot. The proposed input trigger and boundary map are additions to this pilot.

7. Candidate update

Freeze convolutional weights and source BatchNorm running statistics; adapt only BN scale and offset. For binary probabilities p , entropy is $h(p) = -p \log p - (1-p) \log(1-p)$. Compute mean entropy separately within the baseline-predicted foreground and background, then average non-empty strata. This prevents the large background region from dominating the loss, but does not guarantee that either predicted stratum is correct.

The screening pilot uses one SGD step at learning rate $1e-3$. The full study will compare one-step and three-step candidates, a source-snapshot parameter penalty, and a consistency term. For multiple steps, the proposed objective is $L = H_{\text{balanced}} + \lambda_{\text{consistency}} JS(p(x), T^{-1} p(Tx)) + \lambda_{\text{anchor}} \|\phi - \phi_0\|^2$, where ϕ contains adaptable BN affine parameters. Hyperparameters are selected using source validation and prespecified validation corruptions, then locked before held-out testing.

8. Detection, localization and acceptance

First compute a frozen prediction and a transformed prediction. Pixelwise Bernoulli Jensen-Shannon disagreement gives an instability map rather than an explanation of model reasoning. The proposed trigger tests whether its upper quantile exceeds a validation-calibrated limit. A boundary band, derived from the baseline mask, separates edge instability from changes in interiors. Regions can be shown to a reviewer, but cannot be labeled as actual errors without evidence.

After the candidate update, recompute the proxy and predicted foreground area. The pilot accepts only if mean JS decreases by at least $1e-6$ and foreground area changes by at most 0.05 of the input area. The full rule will compare this with boundary-weighted disagreement, relative foreground-area bounds, bounded parameter change and an empty-mask check. An absolute five-point area bound can permit a large relative loss when foreground is sparse. Legitimate appearance or object-size changes can violate these bounds; rejection is conservative behavior, not proof of a bad candidate.

9. Budget and rollback

A hard cap on forward and backward passes sets the budget. Report all candidate and gate costs, including rejected updates. Episodic reset prevents persistent drift between cases; it cannot prevent harm in the current mask. Continual adaptation would require a separate recovery study.

Protocols designed to expose failure

10. Data and partitioning

Stage A starts from the ClinicDB replication checkpoint and the separately trained Massachusetts checkpoint. Use the recorded source frame split for the replication comparison, but use sequence- or patient-disjoint data for later generalization conclusions wherever identifiers allow it. For Massachusetts, retain the original train/validation/test tile split, and assess spatial dependence before interpreting intervals. All crops from a parent tile stay in one partition.

Stage B adds independent polyp acquisition data such as Kvasir-SEG after verifying access and image overlap, and a second geographical or sensor domain for building segmentation. Target labels are reserved for offline scoring. The source-only and supervised target-trained baselines answer different questions and will be reported separately. A new benchmark chosen after inspecting failures requires a new, explicitly labeled evaluation phase.

11. Comparator suite and ablations

Core controls are frozen inference, an identity-update numerical control, flip-based test-time augmentation, inexpensive input normalization, ungated BN-affine adaptation, and the gated version of the same candidate. These establish whether the gate offers value beyond averaging predictions or correcting inputs. Add compatible implementations of Tent, EATA and SAR with clearly documented segmentation adaptations. Where feasible, use the standardized medical protocols of [9] and a parameter-free alternative [8,10]; exclude methods only with a stated resource or architecture reason.

Ablation	Question
No trigger; always form a candidate	Does conditional computation actually save time?
No gate; accept every candidate	How much harm is caused by adaptation itself?
JS only / area only / both	Which signal rejects harmful cases, and which rejects useful ones?
Full-image / boundary-localized proxy	Does localization add value beyond mean disagreement?
1 vs 3 steps; BN affine vs head	Does extra update capacity justify cost and risk?
256/352 inputs vs native crops	Is loss of small structures caused mainly by downsampling?

12. Outcomes and analysis

Report native per-image Dice and IoU, boundary F1 with a documented pixel or physical tolerance, and results stratified by foreground area. For each method, report the rate of Dice losses greater than one point, the mean gain among accepted candidates, harmful-acceptance rate, useful-rejection rate, and accept frequency. Use paired differences against the frozen baseline. Report exact counts, per-case plots and worst failures.

Repeat training with at least three fixed seeds in the full project and use paired bootstrap intervals at the patient/sequence or parent-tile level. The screening pilot uses a single seed and simpler image/tile bootstrap intervals. Measure latency with GPU synchronization and warm-up; report p50/p95 total latency, forward/backward counts, peak allocated and reserved VRAM, and rejection overhead. Synthetic corruptions support a stress-test conclusion, not a clinical or sensor-transfer claim.

Completed runs and a deliberately small pilot

Completed experiment	Native raw Dice	Native raw IoU
ClinicDB; 62 test images	89.59%	83.62%
Buildings; 10 test tiles	71.95%	56.25%

The official architecture has 315,566 parameters. ClinicDB used 200 epochs and mixed precision; Massachusetts used 200 epochs in FP32 at a lower input resolution. These experiments demonstrate execution feasibility, not an adaptation advantage. The source min-max-normalized Dice is 89.59%, compared with the paper's five-run mean of 93.48%. The execution summary documents all material differences.

13. What the adaptation pilot actually tested

The pilot updates 10,872 BN affine parameters for one step, with source statistics frozen, then performs the JS-and-area acceptance test. It resets weights for every image or complete four-patch tile. No additional candidate-learning rate or gate threshold was tuned on the test outcomes. All stress transformations were fixed before the pilot ran.

Condition	Clinic frozen / gated	Buildings frozen / gated
clean	89.59 / 89.57	71.95 / 71.95
brightness_0.65	80.30 / 80.32	59.13 / 59.13
gamma_1.8	63.78 / 63.81	52.50 / 52.50
noise_0.08	24.11 / 24.10	5.34 / 5.34
blur_1.5	88.89 / 88.89	9.64 / 9.65

Values are mean native Dice percentages. Mean candidate overhead was 46.5 ms per polyp image and 56.2 ms per building tile, above the two-forward frozen baseline. Only forward passes were warmed up, so these means include the first adaptation-call cold start. They exclude reset, input corruption and offline native scoring. Per-case records include ungated results, gate decisions and paired bootstrap intervals. The pilot has no input trigger, no learned risk predictor, no boundary-localized gate, and no multi-seed comparison. Those components are proposed; their benefits are not represented as measured.

14. Limits that affect interpretation

Only 62 source images and ten target tiles contribute to these held-out summaries. Nearby source video frames and neighboring aerial tiles can be correlated, so naive image/tile confidence intervals may be too narrow. Downsampled building inputs can remove small objects. Neither dataset establishes clinical usefulness, real edge-device throughput, neuroimaging performance or robustness across institutions. A decrease in flip disagreement cannot certify that a prediction is correct.

The noise condition reduced frozen Dice to 24.11% for polyps and 5.34% for buildings. Blur reduced building Dice to 9.64%, while polyp Dice stayed at 88.89%. The gate accepted 110/310 polyp candidates, including one loss >1 point, but only 1/50 building candidates. This demonstrates a harm-versus-benefit tradeoff, not a successful reliability solution. If harmful candidates remain accepted, the next step will test boundary localization or avoid adaptation on those cases. If the candidate rarely changes predictions, the result motivates a capacity-and-step ablation, not a claim that rollback solved reliability.

Working code and recorded runs

A feasible first research project

15. Milestones and expected outcomes

Period	Concrete milestone and decision
Weeks 1-2	Harmonize reference evaluation; repeat seeds; confirm source/target access and group identities. Release reproducible manifests and leakage checks.
Weeks 3-4	Establish frozen, input-correction, TTA-averaging and ungated-update baselines. Lock validation-derived trigger and gate settings.
Weeks 5-6	Run gate ablations and object-size/boundary analyses. Measure harmful acceptance, useful rejection and synchronized latency.
Weeks 7-8	Test an independent acquisition/geographical domain; stress clean-to-shift-to-clean streams separately from episodic tests.
Weeks 9-10	Repeat key findings; inspect failure cases; package code and write the study. Continue only where gains exceed uncertainty and measured overhead is justified.

Expected artifacts are a compact decision policy, a reproducible evaluation harness, an interpretable map of prediction instability, and evidence about the circumstances in which adaptation is useful or harmful. Publication venue would follow the strength and scope of the results. A negative result would retain value if it identifies a reproducible failure mode and establishes a reliable fallback; a clinical deployment claim would require additional evidence beyond this project.

The work fits experience with constrained-GPU adaptation and error analysis, while adding segmentation, partition-aware evaluation and adaptation decision-making. It aligns with Prof. Rahman's efficiency and reliability pillars without requiring a large new backbone or assuming access to a hospital dataset.

References and working code

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- [12] Authors' ClinicDB training, validation and test split.

Working code and recorded runs

Assistance note. AI tools supported code development, literature organization and initial drafting. All reported measurements come from executed experiments and saved evidence.